

Lab 6: Linux Basics Part 6

Bash Scripting

Quintin Applegate COMP 325-01 FA 2025

Introduction: Bash scripting can be an incredibly powerful tool for cybersecurity experts since it allows them to automate repetitive tasks, deploy security tools quickly, and create their own custom scripts for many endeavors in the cybersecurity space. Some of those may include automating log analysis, monitoring system changes, or processing large datasets. Penetration testers can also use bash scripting to their advantage by creating scripts for reconnaissance automation, exploit development, and post-exploitation tasks. No matter what the task is, bash scripting can save cybersecurity professionals lots of time and hassle. This lab takes a dive into practical bash scripting that emphasizes real-world scenarios where automation is commonly found in security operations

Exercise 1: Basic Bash Structure

```
quintin@kali: ~/c
File Actions Edit View Help
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ chmod +x first_script.sh
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./first_script.sh
=== Security System Check ===
Date: Sun Oct 5 09:39:08 PM EDT 2025
Current user: quintin
Hostname: kali
IP Address: 192.168.1.103
=== Check Complete ===
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ vim script_template.sh
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ chmod +x script_template.sh
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./script_template.sh
=====
Security Template v1.0
Started: Sun Oct 5 09:43:08 PM EDT 2025
=====
Script is running... Press Esc. then type :wq to save and exit
=====
Script completed: Sun Oct 5 09:43:08 PM EDT 2025
=====
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$
```

Exercise 2: Variables and User Input

```
quintin@kali: ~/cybersec-labs/bash-scripting-lab
Cybersecurity Lab Series - Lab6.pdf — Mozilla Firefox
File Actions Edit View Help
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ nano variables_demo.sh
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ chmod +x variables_demo.sh
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./variables_demo.sh
=== Scan Configuration ===
Target IP: 192.168.1.100
Scan Type: tcp
Port Range: 1-1000
Timeout: 30 seconds
Output Directory: /tmp/scans

=== System Information ===
Date: 20251006
User: quintin
Uptime: 11:20:23 up 1:51, 2 users, 0 load average: 0.10, 0.08, 0.04
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$
```

```
quintin@kali: ~/cybersec-labs
File Actions Edit View Help
User: quintin
Uptime: 11:20:23 up 1:51, 2 users, load average: 0.10, 0.08, 0.04

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ vim user_input_demo.sh

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ chmod +x user_input_demo.sh

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./user_input_demo.sh
== Security Scan Configuration ==

Enter target IP address: 192.168.1.0
Enter scan type (tcp/udp): tcp
Enter output filename: user_input_outpt.txt

== Configuration Summary ==
Target: 192.168.1.0
Scan Type: tcp
Output File: user_input_outpt.txt

Proceed with scan? (y/n): y
Starting scan...
Scan would execute here with parameters:
  Target: 192.168.1.0
  Type: tcp
  Output: user_input_outpt.txt

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$
```

Exercise 3: Conditional Statements

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]_emo.sh
$ ./case_demo.sh
Security Scan Menu
1) Network Discovery
2) Port Scanning
3) Service Enumeration
4) Vulnerability Scanning
5) Exit

Select scan type (1-5): 2
Starting Port Scanning...
Command: nmap -sS -p 1-1000 [target]e_demo.sh

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ nmap -sS -p 1-1000 443
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-06 12:20 EDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.17 seconds

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$
```

Exercise 4: Loops for Automation

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
```

```
$ vim while_loop_demo.sh
```

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
```

```
$ chmod +x while_loop_demo.sh
```

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
```

```
$ ./while_loop_demo.sh
```

```
=== Security Service Monitor ===
```

```
Monitoring ssh for 30 seconds ...
```

```
Press Ctrl+C to stop monitoring
```

```
Check #1 - 22:55:58 - [!]
```

```
ssh: STOPPED
```

```
Check #2 - 22:56:03 - [!]
```

```
ssh: STOPPED
```

```
Check #3 - 22:56:08 - [!]
```

```
ssh: STOPPED
```

```
Check #4 - 22:56:13 - [!]
```

```
ssh: STOPPED
```

```
Check #5 - 22:56:18 - [!]
```

```
ssh: STOPPED
```

```
Check #6 - 22:56:23 - [!] done
```

```
ssh: STOPPED
```

```
Monitoring completed after 30 seconds Total checks performed: 6
```

```
Total checks performed: 6
```

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
```

```
$ █
```

Press Esc, type :wq

Exercise 5: Functions for Code Organization


```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ nano functions_demo.sh

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./functions_demo.sh
[*] Security Assessment Starting
[*] Starting host discovery for 127.0.0.1
[+] 127.0.0.1 is reachable
[!] No common ports found open
[*] Generating report: scan_report_20251006_232223.txt
[+] Report saved to scan_report_20251006_232223.txt
[+] Security Assessment Complete

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$
```

Exercise 6: File Processing and Log Analysis

```
(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./log_analyzer.sh
Starting security log analysis...
./log_analyzer.sh: line 65: [: 0
0: syntax error in expression (error token is "0")
Analysis complete. Report saved to: security_analysis_20251007_012335.txt
=== Authentication Log Analysis ===
Analysis Date: Tue Oct 7 01:23:35 AM EDT 2025
Press Esc, type :wq to save and exit

Successful logins: 0
Failed login attempts: 0

Potential brute force sources:
Privilege escalation attempts:
Invalid user login attempts:

=== Security Recommendations ===
- Disable root login via SSH
- Use sudo for administrative tasks instead of root login
14
```

Exercise 7: Network Security Automation

```

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./network_recon.sh -n "192.168.1.0/24"
== Network Reconnaissance Script ==
Target network: 192.168.1.0/24
Output directory: ./recon_results
Timestamp: 20251007_014756

[*] Starting host discovery for 192.168.1.0/24
[+] Host discovery completed. Results saved to: ./recon_results/host_discovery_20251007_014756.txt
[*] Starting port scanning
[+] Port scanning completed. Results saved to: ./recon_results/port_scan_20251007_014756.txt
[+] Summary report generated: ./recon_results/reconnaissance_summary_20251007_014756.txt

== Reconnaissance Complete ==
Check the output directory for detailed results: ./recon_results

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$

```

Exercise 8: System Security Monitoring

```

(quintin@kali)-[~/cybersec-labs/bash-scripting-lab]
$ ./security_monitor.sh --check
2025-10-07 20:32:10 [INFO] Starting security monitoring cycle
2025-10-07 20:32:10 [ALERT] High number of failed login attempts detected: 6
2025-10-07 20:32:10 [ALERT] IP 203.0.113.51: 3 failed attempts
2025-10-07 20:32:10 [ALERT] IP 203.0.113.50: 3 failed attempts
2025-10-07 20:32:10 [ALERT] Suspicious process detected: nc
2025-10-07 20:32:11 [INFO] Security monitoring cycle completed

```

Knowledge Assessment:

1. True
2. A
3. True
4. A
5. B

Conclusion: This lab demonstrated how bash scripting can streamline cybersecurity operations through automation, efficiency, and customization. By learning to create and execute bash scripts, I gained practical experience in building tools that can automate reconnaissance, monitor systems, and analyze logs. All skills that are crucial in real-world security environments. These scripting techniques not only save time but also reduce human error and increase consistency across tasks. With these fundamentals, I am now better equipped to integrate automation into security operations, incident response, and

penetration testing, making my work more effective and resourceful as an aspiring cybersecurity professional.